Approach 1 — TF-IDF: Lexical Distinctiveness, interpretive questions

Figure 1: set of tables showing distinctive terms by document

1. **Do some documents share distinctive vocabulary?**
   Several documents share overlapping distinctive vocabulary, which points to genuine thematic clusters within the corpus. For example, texts 9 and 13 share almost identical top terms like

"plantations," "abatement," "cent," "poor," "usury," "interest," "newfoundland," "2dly," "pag," "barbadoes" indicating that they are closely related. Text 14 and 18 similarly share multiple top words like "poor," "parish," "fathers," "propose," "hospitals," "assembly," "election," "elected," and "communication." Given that Tf-idf is a measure of relative distinctiveness and still reveals shared vocabulary across these pairs tells us the overlap is strong enough to survive the penalty for common corpus-wide terms.

Are distinctive terms topical, rhetorical, or technical?

The distinctive terms are predominantly topical and technical rather than rhetorical. For example, the monetary cluster (texts 1, 4, 6, 19) is dominated by technical financial vocabulary like "exchange," "stivers," "ducats," "fineness," "doller," "undervaluation." whereas another cluster of texts 9, 13 and 15 is heavy on colonial terminology like "plantations," "barbadoes," "usury," "cent," "newfoundland."

Approach 2 — Pearson Correlation: Similarity and Distance

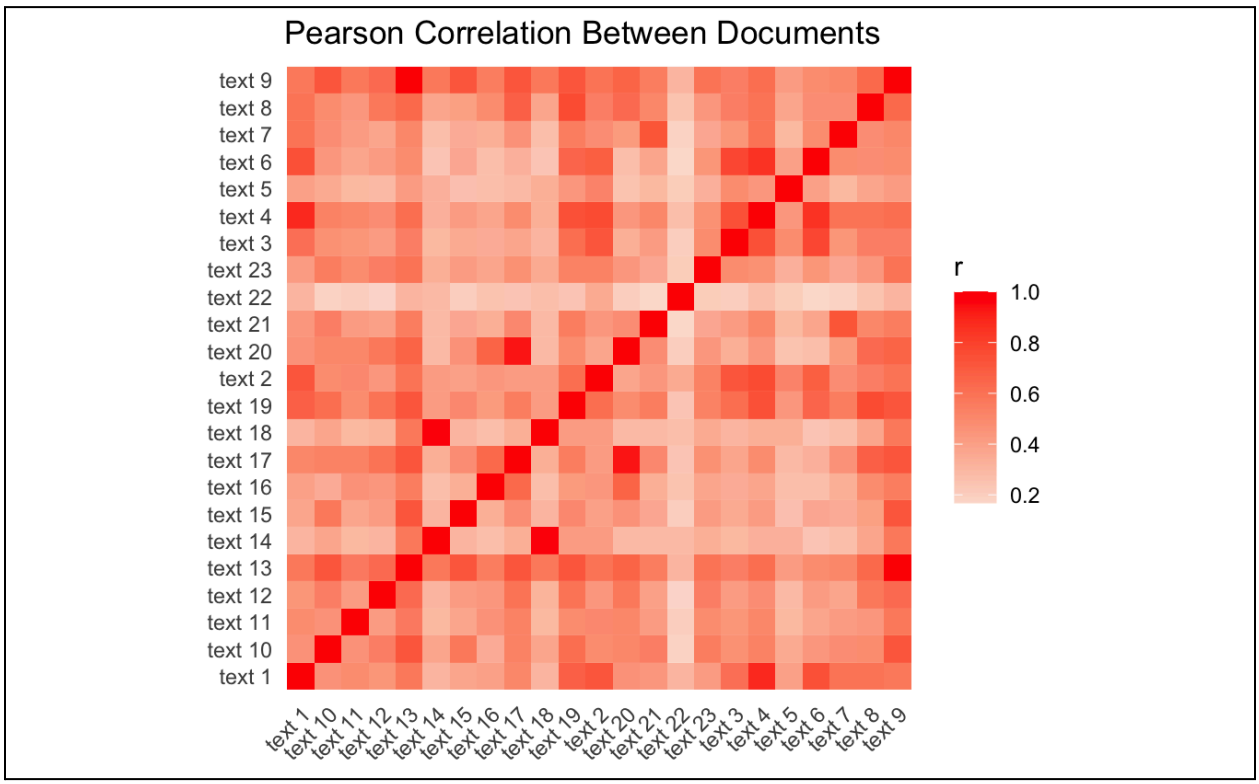

Figure 2: similarity heatmap

Two most similar pairs

The most similar pair is text 13 and text 9, with a Pearson correlation of exactly 1.0. This is essentially a perfect correlation, meaning these two documents have an identical pattern of word use across the entire vocabulary space. Combined with their near-identical TF-IDF profiles (both topped by "plantations," "abatement," "barbadoes," "usury," "newfoundland"), a correlation of 1.0 strongly suggests these are not independent texts but could be reprints or separate editions of the same writing. The second most similar pair is text 14 and text 18, with r = 0.983. This is not a

perfect correlation, so they are distinct documents, but the near-perfect score combined with their shared TF-IDF vocabulary ("poor," "parish," "fathers," "hospitals," "assembly," "election," "elected") suggests they are closely related.

Two least similar pairs

The least similar pair is text 22 and text 6, with r = 0.168, followed closely by text 21 and text 22 at r = 0.169. Both pairings include text 22. Notably these correlations are positive rather than negative, meaning there is still some baseline shared vocabulary, but the overlap is minimal.

What questions does this pattern generate?

Three questions follow directly from these numbers. First, should text 9 and text 13 be treated as one document for the rest of the analysis? A correlation of exactly 1.0 means they are computationally identical and keeping both inflates the apparent size of the colonial/plantation cluster and could distort any similarity measures that include them.

Second, text 22's consistent low correlation with everything raises the question of whether it belongs in this corpus at all. If it is a legal document about a property dispute rather than a political economy text, its presence may reflect an archival categorization decision worth questioning.

Third, the fact that the least similar pairs all involve text 22 rather than text 23 (Adam Smith) is itself surprising, suggesting that lexical distance in this corpus is driven more by subject matter specificity than by historical period.

Approach 3 — Syntactic Complexity

doc_id <chr>	MLS <dbl>	C_per_S <dbl>	DepCl_per_S <dbl>	Coord_per_S <dbl>	ComplexNom_per_S <dbl>
wealth	167.87	15.11	8.77	6.92	28.10
text 7	490.86	45.26	26.69	26.47	85.29

2 rows

Figure 3: syntax table

How do the two texts differ in syntactic complexity?

Text 23: "in every other art and manufacture the effects of the division of labor are similar to what they are in this very trifling one though in many of them the labor can neither be so much subdivided nor reduced to so great a simplicity of operation..."

Text 7 : "the circle of commerce... and surely my selfe being conscious of mine owne wants yet as sensible as any of the terrors without and errors within wherewith the commerce of this kingdome is encountered layd mine hand on my mouth and kept long silence because it was a

subiect fit for a more able man... but when i beheld this former flourishing trade of ours to be threatened by many as eminent as imminent dangers..."

These two sentences illustrate the key syntactic difference well. Text 23 builds its argument by introducing a concept (division of labor), and then systematically breaking it down into sub-points, with each clause adding a specific qualification or example. It is long but feels organized. Text 7 by contrast jumps between a classical story, a personal confession, and a political warning all within one sentence, joining everything together with "and," "but," and "whilst." This matches exactly what the syntactic measures showed: text 7 has three times as many clauses per sentence (45.26 vs 15.11), three times as many dependent clauses per sentence (26.69 vs 8.77), nearly four times as much coordination per sentence (26.47 vs 6.92), and three times as many complex nominals per sentence (85.29 vs 28.10). The consistency of this difference across all five measures indicates a fundamentally different approach to sentence construction between the two texts.

Do these differences align with or complicate your earlier lexical findings?
They largely align with my earlier findings since the TF-IDF and Pearson results already established that text 23 and text 7 are among the most lexically distant documents in the corpus. The syntactic data reinforces this by showing they also write differently at the structural level, not just the word level.

What rhetorical or stylistic practices might these patterns reflect?
Based on the syntactic measures alone, text 7 appears to use very long, complex sentences with lots of clauses joined together. Text 23, on the other hand, has shorter sentences with more embedded clauses, which suggests a more structured and organized writing style. The numbers suggest that text 7 relies more on building up arguments through length and repetition, while text 23 seems more focused on presenting ideas in a cleaner, more step-by-step way. These differences in style likely reflect the different time periods and purposes of the two texts.

Synthesis

After looking at all three approaches together, the central question I want to ask is whether this text corpus contains two distinct groups of texts and if quantitative methods detect the boundary between them.

The TF-IDF results gave the first hint of two distinctive groupings: most texts share vocabulary around practical, policy-focused words like "exchange," "realm," "moneys," "commodities," and "plantations." Text 23 stands completely apart with top terms like "labor," "capital," "revenue," and "system," which are much more abstract. This suggested early on that text 23 might belong to a different kind of writing than the rest of the corpus.

The Pearson correlation confirmed this. Text 23 shows consistently pale coloring against almost every other document in the heatmap, meaning it shares very little vocabulary pattern with the rest of the corpus. What was surprising is that even texts that seem like they should be related to Text 23 (Adam Smith), like the monetary texts 1, 4, and 6, still show low correlation with text 23. This suggests the vocabulary gap between these presumably older texts (since I have not read any of them besides Smith) and Smith is larger than expected.
The syntactic complexity results added another layer. Text 7 had nearly three times longer sentences and much higher coordination than text 23 on every single measure. I initially expected text 23 to be more syntactically complex since Smith is building a more elaborate theoretical system, but the opposite was true. This suggests that syntactic complexity in this corpus reflects writing style more than intellectual complexity. All three methods point in the same direction: text 23 is genuinely different from the rest of the corpus, and that difference shows up in vocabulary, sentence structure, and word patterns.

However, it is important to acknowledge what these methods cannot tell us. TF-IDF and Pearson correlation only measure word frequencies and cannot tell us whether the distance between Smith and the pamphlets reflects a real intellectual break or just a natural change in vocabulary over time. The syntactic parser was also trained on modern English, which means it may not be reading Early Modern sentences in text 7 correctly, potentially inflating its complexity scores. So while all three methods point toward the same pattern, each has limitations, and the conclusion should be treated as a starting point for further research rather than a definitive answer.